Conditional Normalizing Flow for Fast Generation of MC SMEFT Events

A case study on ttZ data generation conditioned on Wilson coefficient cHt

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Outline

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Motivation

- Current MC simulation slow
- Need for fast generation of events
- Potential of normalizing flows
- Method widely applicable to other processes and parameters

Motivation: MC simulation slow

- In 2017 20% of the ATLAS computing resources were used for MC simulation [1]
- Current MC generation can take days or weeks for complex processes
- With the HL-LHC, more complex events and more data will require even more MC simulation
- We need faster and more efficient MC generation

Outline

Idea

- Take already existing MC data
- Train a conditional normalizing flow to learn the distribution of events
- Condition the flow on a Wilson coefficient
- Use the trained flow to generate new events for any value of the Wilson coefficient, much faster than traditional MC simulation

Outline

Normalizing flow: MAFMADEMOG

- Masked Autoregressive Flow (MAF)
  - A type of normalizing flow that models complex distributions by transforming a simple base distribution through a series of invertible transformations
- MADE (Masked Autoencoder for Distribution Estimation)
  - A type of autoregressive model that allows for efficient sampling and density estimation
- MOG (Mixture of Gaussians)
  - Used as the base distribution for the flow, allowing it to model complex distributions

Interactive Architecture Artifact

- Click to open the interactive visualization:

Open Claude Artifact (interactive)

`https://claude.ai/public/artifacts/72c0654f-7dad-4cef-ac68-80df06001473`

Equation Example

$$p(x \mid \theta) = \frac{1}{Z(\theta)} \exp\left(-E(x;\theta)\right)$$

Outline

Figure Placeholder

Insert figure here

Table Example

Model	Metric A	Metric B
Baseline	0.91	0.78
Improved	0.94	0.83

Outline

Takeaways

- Main conclusion
- Supporting evidence
- Next steps

Backup

Questions?

References I

The HSF Physics Event Generator Working Group, Andrea Valassi, Efe Yazgan, Josh McFayden, Simone Amoroso, Joshua Bendavid, Andy Buckley, Matteo Cacciari, Taylor Childers, and Vitaliano Ciulli.
Challenges in monte carlo event generator software for high-luminosity lhc.
arXiv:2004.13687 [hep-ph], February 2021.